

Azure Monitoring and Identity Management

Project Overview

This document summarizes hands-on experience configuring monitoring, alerting, and identity management in Microsoft Azure using Azure Monitor and Microsoft Entra ID.

Monitoring and Alerting with Azure Monitor

This section documents the configuration of Azure monitoring and alerting capabilities, including activity log review, alert creation, and exploration of Azure Monitor tools for operational visibility.

Azure Activity Log Review

The screenshot displays the Microsoft Azure portal's Activity Log. The left sidebar shows the navigation menu with 'Monitor | Activity log' selected. The main content area shows the 'Update Storage Account Create' event for the resource 'AZ900RGlod58896338'. The event details include the change type 'Create', the client type 'PowerShell', and the changed by user 'e948cdc5-a8c8-49e8-a9...'. The right sidebar shows a 'Check your work' section with a checklist item 'Verify that you have reviewed the Azure Monitor activity log page' which is marked as complete.

Change Time	Changes	Change Type	Changed By	Client Type
Thursday, February 5, 2026 14:41:58 GMT-0500 (Eastern Standard Time)	Resource Created	Create	e948cdc5-a8c8-49e8-a9...	PowerShell

Alert Creation and Configuration

The screenshot shows the Microsoft Azure portal interface. The main area displays the 'Action groups' page for the 'Challenge Labs 06' subscription. A table lists one action group: 'Virtual Machine Admins' (Short name: VM Admins, Resource group: az900rglod58896338, Subscription: Challenge Labs 06, Actions: 1 Email, Status: Enabled). The right-hand side panel, titled 'AZ900-010: Configure Monitoring by using Azure Monitor', provides instructions for creating an alert. It includes a 'Hints Enabled' toggle (set to 'No') and a list of steps: 1. Create an action group named 'Virtual Machine Admins' using the 'AZ900RGlod58896338' resource group, a short name of 'VM Admins', and an 'Email/SMS message/Push/Voice' notification named 'VM Shutdown' that will send an email to your personal email account when the action is triggered. 2. Verify that your personal email address has been added to the 'VM Shutdown' action group. 3. Create an activity log alert rule named 'VM1 Shutdown' that will send an 'Informational' alert with the description 'VM1 has shut down to'.

Name	Short name	Resource group	Subscription	Actions	Status
Virtual Machine Admins	VM Admins	az900rglod58896338	Challenge Labs 06	1 Email	Enabled

The email notification is from Microsoft Azure (azure-noreply@microsoft.com) to the user. It states: 'You're now in the VM Admins action group'. The main body of the email says: 'You've been added to an Azure Monitor action group. You are now in the VM Admins action group and will receive notifications sent to the group. View details on Azure Monitor action groups >'. Below this, under the heading 'Account information', the following details are provided: Subscription ID: D28DF0A3-5E54-471F-B9CE-DF15FA1A8FF3, Resource group name: AZ900RGlod58896338, and Action group name: Virtual Machine Admins.

You're now in the VM Admins action group

Microsoft Azure <azure-noreply@microsoft.com>
To: You
Thu 2/5/2026 2:49 PM

Microsoft Azure

You've been added to an Azure Monitor action group

You are now in the VM Admins action group and will receive notifications sent to the group.

[View details on Azure Monitor action groups >](#)

Account information

Subscription ID: D28DF0A3-5E54-471F-B9CE-DF15FA1A8FF3

Resource group name: AZ900RGlod58896338

Action group name: Virtual Machine Admins

Alert rules - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/AlertRulesBlade/subscriptions-~/%5B%22d28df0a3-5e54-471f-b9ce-df15fa1a8ff3%22%5D

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Admin: 58896338@cloud...
CLOUDSUCKE (CLOUDSUCKE.ONLINE)

Home > Monitor | Alerts

Alert rules

+ Create | Columns | Refresh | Export to CSV | Open query | Delete | Enable | Disable

Search

Subscription: Challenge Labs 06

Target scope: all

Add tag filter

More (4)

No grouping

Name	Condition	Severity	Target scope	Target resource type	Signal type	Status
VM1 Shutdown	Category=Administrati...	4 - Verbose	VM1	Virtual machine	Activity log	Enabled

Showing 1 - 1 of 1 results.

Give feedback

AZ900-010: Configure Monitoring by...

labclient.labondemand.com/Instructions/3c...

AZ900-010: Configure Monitoring by Usi...
22 Minutes Remaining

Instructions | Resources | Help

100%

- Create an action group named **Virtual Machine Admins** by using the **AZ900RGlod58896338** resource group, a short name of **VM Admins**, and a **Email/SMS message/Push/Voice** notification named **VM Shutdown** that will send an **email** to your personal email account when the action is triggered.
- Verify that your personal email address has been added to the **VM Shutdown** action group.
- Create an activity log alert rule named **VM1 Shutdown** that will send an **Informational** alert with the description **VM1 has shut down** to the **Virtual Machine Admins** action group when the **Deallocate Virtual Machine (Virtual Machines)** signal is triggered by **VM1**.
- Verify that the alert rule named **VM1 Shutdown** exists and that the status is **enabled**.

Previous | Next: Explore the Azure...

Azure Monitor Service Overview

Monitor - Microsoft Azure - Google Chrome

portal.azure.com/#view/Microsoft_Azure_Monitoring/AzureMonitoringBrowseBlade/~-/metrics

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Admin: 58896338@cloud...
CLOUDSUCKE (CLOUDSUCKE.ONLINE)

Home > Monitor

Monitor | Metrics

What metrics are available for this resource? | What metrics do people commonly use to track this kind of resource? | +1

Search

+ New chart | Refresh | Share

Local Time: Last 24 hours (Automatic)

Chart Title

+ Add metric | Add filter | Line chart | Drill into Logs | New alert rule

Apply splitting | Save to dashboard

Scope	Metric Namespace	Metric	Aggregation
VM1	Virtual Machine Host	Select metric	Select aggregation

Select a metric above to see data appear on this chart or learn more below:

Filter + Split

Plot multiple metrics

Build custom dashboards

6 PM | Thu 05 | 6 AM | 12 PM | UTC-05:00

Add or remove favorites by pressing Ctrl+L+Shift+F

Give Feedback

AZ900-010: Configure Monitoring by...

labclient.labondemand.com/Instructions/3c...

AZ900-010: Configure Monitoring by Usi...
20 Minutes Remaining

Instructions | Resources | Help

100%

- Display the contents of the **Metrics** page on the Azure Monitor service menu, and then review metrics for **VM1**.
- You will not see any data on the Metrics page because you have not configured VM1 for diagnostics and data collection.
- Display the contents of the **Logs** page on the Azure Monitor service menu.
- If prompted to select a scope, select the **AZ900RGlod58896338** resource group.
- In this Challenge Lab, you will not be able to run a query because a Log Analytics workspace has not been created for your subscription.
- Display each of the remaining Azure

Previous | Next

Identity and Group Management with Microsoft Entra ID

This section documents the management of users and groups using Microsoft Entra ID, focusing on identity organization and access control within an Azure environment.

Modified the Technical Contact for the Microsoft Entra ID tenant

The screenshot shows the Microsoft Entra ID Properties page for the tenant Hexelo40898132x2. The Technical contact field is updated to LabAdmin@hexelo.com. The page also displays various other tenant properties such as Name, Tenant Country/Region, Data location, Notification language, and Tenant ID.

Hexelo40898132x2 | Properties

- Name: Hexelo40898132x2
- Tenant Country/Region: United States
- Data location (Primary Copy): United States datacenters
- Notification language: English
- Tenant ID: fff02731-e768-475c-859b-519948bc2613
- Technical contact: LabAdmin@hexelo.com
- Global privacy contact:
- Privacy statement URL:

[Learn more](#) about data location and residency.

Access management for Azure resources

(LabAdmin@hexelo40898132x2.onmicrosoft.com) can manage access to all Azure subscriptions and management groups in this tenant. [Learn more](#)

Save Discard

Check your work

- Verify that you have modified the Technical Contact for the Microsoft Entra ID tenant.

Congratulations! You have modified the Technical Contact for the Microsoft Entra ID tenant.

Verify

Next: Create a Microsoft Entra...

Microsoft Entra ID user management interface displaying configured users

The screenshot shows the Microsoft Entra ID Users page for the tenant Hexelo40898132x2. The page displays a list of users with columns for Display name, User principal name, User type, Is Agent, and On-premises sync. The Riley Burgess user is highlighted.

Users

Hexelo40898132x2

[New user](#) [Edit](#) [Delete](#) [Download users \(Preview\)](#) [Bulk operations](#) [Refresh](#)

[Azure Active Directory is now Microsoft Entra ID.](#)

4 users found (1 user selected)

☐	Display name ↑	User principal name ↴	User type	Is Agent	On-premises sync
☐	admin	admin@hexelo40...	Member	No	No
☐	LabAdmin	LabAdmin@hexel...	Member	No	No
☐	Office365Admin	Office365Admin@...	Member	No	No
☒	Riley Burgess	RileyB@Hexelo40...	Member	No	No

Check your work

- Verify that you have created a new Microsoft Entra user account named Riley Burgess.

Congratulations! You have created a new Microsoft Entra user account named Riley Burgess.

Verify

Next: Create a Microsoft Entra...

Group configuration and membership settings in Microsoft Entra ID

The screenshot displays the Microsoft Azure portal interface. The main window shows the 'Groups' section with a list of groups. A search bar and filters are visible. A table lists the groups found:

Name	Object id	Group type
MyGroup	5b66857c-1676-4147-be6c-7c040e2ef84	Microsoft 365

On the right, a guided tour for 'AZ900-013: Manage Microsoft Entra Users and Groups' is shown. It includes instructions, resources, and a 'Check your work' section with verification steps:

- Verify that you have created a Microsoft Entra group named MyGroup. Congratulations! You have created a Microsoft Entra group named MyGroup.
- Verify that you have added a user named Riley Burgess to the Microsoft Entra group named MyGroup. Congratulations! You have added a user named Riley Burgess to the Microsoft Entra group named MyGroup.

A 'Verify' button is present at the bottom of the tour.

User password reset and account management action in Microsoft Entra ID.

The screenshot shows the 'AZ900-013: Manage Microsoft Entra Users and Groups [Guided]' page. The 'Instructions' tab is active, displaying the following content:

Change a user's password during initial sign in

Hints Enabled
No ☐ Yes ☒

- Open an incognito or InPrivate browser window, go to <https://portal.azure.com>, sign in as RileyB@hexelo40898132x2.onmicrosoft.com using [InitialPwd58429122](#) as the password, and then when prompted, change the password to [NewPwd58429122](#).

Check your work

- Verify that you have signed in to the Azure portal as Riley Burgess and reset the user password. Congratulations! You have signed in to the Azure portal as Riley Burgess and reset the user password.

A 'Verify' button is located at the bottom of the instructions.